ANSWER KEY & MARKING SCHEME SET A

Chapters 11 & 12: **Exploring Some Geometric Themes, Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 27	Each stage multiplies the count by 3: $3^3 = 27$
2	**(a)** 6	Count the faces
3	**(c)** 5	Count the faces
4	**(a)** 4	Count the faces
5	**(d)** 12	Euler's relation
6	**(d)** 46.4286	Sum 325 ÷ 7
7	**(b)** 140	Sum 700 ÷ 5
8	**(c)** 26	Arrange in order and take the middle value
9	**(c)** 12	Arrange in order and take the middle value
10	**(d)** 30	Total = mean × n

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A part of the figure looks like the whole figure (2).	2
12	It becomes $\frac{4}{3}$ times the previous length (2).	2
13	Total 54 then 70; number 16 (2).	2
14	On a number line with dots as weights, the mean is where the line balances (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Surface area $= 6 \times 2^2 = 24$ cm² (2); 2 cubes (1).	2+1
16	Total $= 160$ (1); sum of known $= 130$ (1); missing value $= 30$ (1).	1+1+1
17	Surface area $= 6 \times 3^2 = 54$ cm² (2); 2 cubes (1).	2+1
	OR: Surface area $= 6 \times 5^2 = 150$ cm² (2); 2 cubes (1).	2+1
18	Total $= 72$ (1); sum of known $= 60$ (1); missing value $= 12$ (1).	1+1+1
	OR: Total $= 200$ (1); sum of known $= 150$ (1); missing value $= 50$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 2 cm (2). (ii) 27 (1). (iii) $(3/4)^3 = 27/64$ (2).	2+1+2
	OR: (i) Hexagonal prism, 8 faces, 18 edges (3). (ii) $8 + 12 - 18 = 2$ (2).	3+2
20	(i) 42 (2). (ii) For 17 numbers, the median is the 9th value (3).	2+3
	OR: (i) Both are multiplied by 2 (3). (ii) Mean 5 to 10, median 5 to 10 (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 11 & 12: **Exploring Some Geometric Themes, Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** fractal	Self-similarity
2	**(b)** 6	6 faces
3	**(d)** 4	One per base edge
4	**(a)** triangle	Looking from the side
5	**(a)** $F + V - E = 2$	Euler's formula
6	**(a)** change over time	Trend
7	**(c)** range	Spread
8	**(c)** median	Median
9	**(b)** 10	All equal
10	**(b)** 40	5 × 8

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A circle (2).	2
12	$12 \times 4 = 48$ cm (2).	2
13	No: mean of 2 and 3 is 2.5 (2).	2
14	The median also increases by 10 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Surface area $= 6 \times 6^2 = 216$ cm² (2); 2 cubes (1).	2+1
16	Greatest rise is between April and May (a rise of 7) (1.5); average $= 138/5 = 27.6$ (1.5).	1½+1½
17	Cube net has six equal squares; cuboid net has rectangles of three pairs (3).	3
	OR: Start with an equilateral triangle (1); divide each side into three equal parts and add a small triangle on the middle part (1); repeat on every new side (1).	1+1+1
18	Total 360; remaining 308 (2); left boy 52 kg (1).	2+1
	OR: Dots stacked above each value (2); 4 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Prisms have two bases and rectangles; pyramids one base and triangles (3). (ii) Four triangles, one central (2).	3+2
	OR: (i) 5, 9, 6 (2). (ii) $6 \times 4 + 3 \times 20 = 84$ cm (2). (iii) Triangle (1).	2+2+1
20	(i) Both 74 (2). (ii) A (1). (iii) 8 and 58 (2).	2+1+2
	OR: (i) $x = 8$ (2). (ii) 7.2 (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 11 & 12: **Exploring Some Geometric Themes, Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 81	Each stage multiplies the count by 3: $3^4 = 81$
2	**(d)** 12	Count the edges
3	**(a)** 9	Count the edges
4	**(d)** 6	Count the edges
5	**(d)** 9	Euler's relation
6	**(b)** 11.6667	Sum 70 ÷ 6
7	**(d)** 48.3333	Sum 290 ÷ 6
8	**(b)** 9	Arrange in order and take the middle value
9	**(a)** 21	Mean of the two middle values
10	**(b)** 12	Total = mean × n

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	10 (2).	2
12	Fern leaf, broccoli, river networks, snowflakes (1+1).	1+1
13	Both 50 (2).	2
14	Values far from the rest (1); 54 in 16, 18, 20, 22, 54 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Segments multiply by 4 each stage (1); $4^3 = 64$ segments (2).	1+2
16	Mean 5.5 (1.5); median 5.5 (1.5).	1½+1½
17	12 edge cubes (3).	3
	OR: $2(40 + 24 + 15) = 158$ cm² (3).	3
18	New mean 15 (1.5); new total 75 (1.5).	1½+1½
	OR: Bar graph compares categories (1.5); line graph shows trend over time (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 64 (2). (ii) 8 corner cubes (3).	2+3
	OR: (i) $V = 2 - F + E = 8$ (3). (ii) Cube and cuboid (2).	3+2
20	(i) Axes, plotted points, trend (3). (ii) Temperature, stock prices (2).	3+2
	OR: (i) Mean 8; deviations $-4, -2, 0, 2, 4$ sum 0 (3). (ii) Left and right deviations balance (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET D

Chapters 11 & 12: **Exploring Some Geometric Themes, Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the middle third of each side	Fractal construction
2	**(a)** cone	Cone
3	**(b)** square	Same as a face
4	**(c)** the central inverted triangle	Fractal
5	**(d)** prism	Definition
6	**(a)** increases by 5	Shift
7	**(d)** a rapid increase	Slope
8	**(a)** 150	10 × 15
9	**(b)** 6	30 ÷ 5
10	**(c)** 5	7 − 2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sphere (2).	2
12	A square (2).	2
13	$10/6 \approx 1.67$°C per hour (2).	2
14	Mean $x + 2 = 10$; $x = 8$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	One base and seven triangles (3).	3
16	$x = 9$ (1); mean $47/5 = 9.4$ (2).	1+2
17	Square, circle, circle with centre, square with diagonals (3).	3
	OR: Packaging boxes, paper models and gift boxes (1.5); designing 3D objects from flat sheets (1.5).	1½+1½
18	Mean $(a + b)/2$ is halfway (3).	3
	OR: Both means 67.5 and 67.5 (1.5); ranges 15 and 35 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Repeating patterns at all scales (3). (ii) Fern leaf, Romanesco broccoli, kolam designs (2).	3+2
	OR: (i) 4 triangles and 1 square (1). (ii) $\sqrt{20^2 + 30^2} = \sqrt{1300} \approx 36$ m (2). (iii) $\frac{1}{2} \times 40 \times 36 \approx 720$ m² (2).	1+2+2
20	(i) Mean 38.75; median $(38 + 45)/2 = 41.5$ (2). (ii) 90 (1). (iii) Large variation (2).	2+1+2
	OR: (i) $4 \times 12 = 48$; $48 - 32 = 16$ (3). (ii) Sorted 8, 10, 14, 16; median $(10 + 14)/2 = 12$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.